INVESTMENTS 1005 / GROUP 9

A-share cross-sectional stock selection

Fundamentals, market signals and machine learning

2 Jan 2018 – 29 Apr 2026
Daily inputs · Top 100 · 20-session targets

One comparison, three periods

The same published model versions in the tables, curves and conclusions

Model version	Full net	2021–2025 net	2026 net
Ridge-exp	395.35%	195.66%	8.80%
Lasso-3y	417.56%	211.46%	3.35%

Ridge-exp: original expanding history · Lasso-3y: optimized three-year window

Full history includes 2018–2020 fitted replay; 2026 ends on 29 April.

Ridge-3y control is a separate development diagnostic, absent from the main curves.

Lasso-3y leads full/development; Ridge-exp leads 2026.

Why combine these signals?

Fourteen candidate inputs; 12 used across the frozen models

Fundamentals

Profitability and cash generation

Signed book and earnings yields

Growth and leverage dynamics

F1 · F2 · F3 · F5 · F6 · F7 · F8 · F9 · F10

Market behaviour

20- and 60-session momentum

126-to-20-session momentum

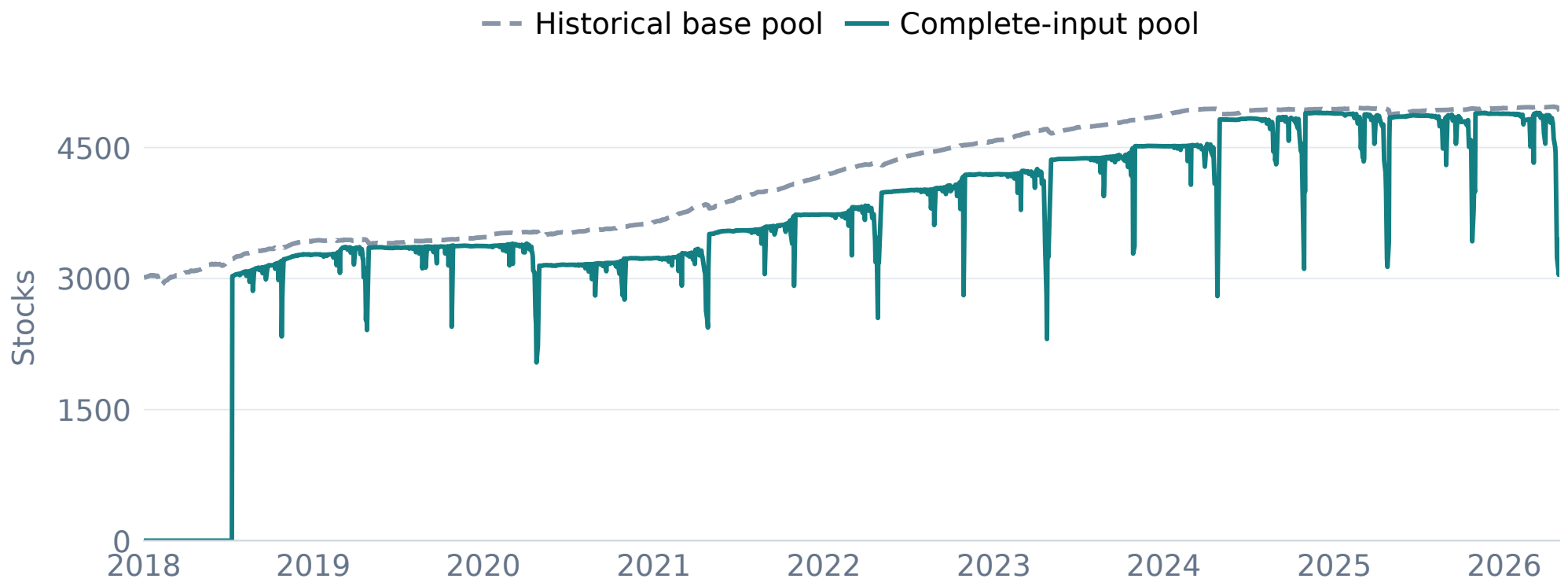
Return volatility and turnover

T1 · T2 · T3 · T5 · T6

Distinct mechanisms motivate the inputs; their usefulness is empirical.

Data and the common stock pool

2,018 sessions · 6,000 historical slots · First signal: 11 Jul 2018



The same fourteen-input finite pool is used by every portfolio.

Fixed models, annual refits

Predict 20-session returns; only labels with earlier t+21 exits enter training

Input ranks → **Annual model** → **Style residuals** → **Causal EMA** → **Score ranks**

Method	Score or model	EMA sessions
Fundamental	Signed financial ranks	20
Combined	Equal financial / technical families	40
Ridge-exp	**Expanding history; alpha = 10,000**	**20**
Lasso-3y	8 inputs; 3-year history; validation alpha	20
LightGBM	depth 3; 7 leaves; 56 trees	40

Industry and log-size residualization precede the causal score EMA.

Portfolio construction and execution

Top 100 and equal-weight pool portfolios use the same implementation rules

Selection and timing

Equal-weight Top 100
Continuous 20-session targets
Close signals, next-open trading
T+1, limits and order retries

Matched benchmark

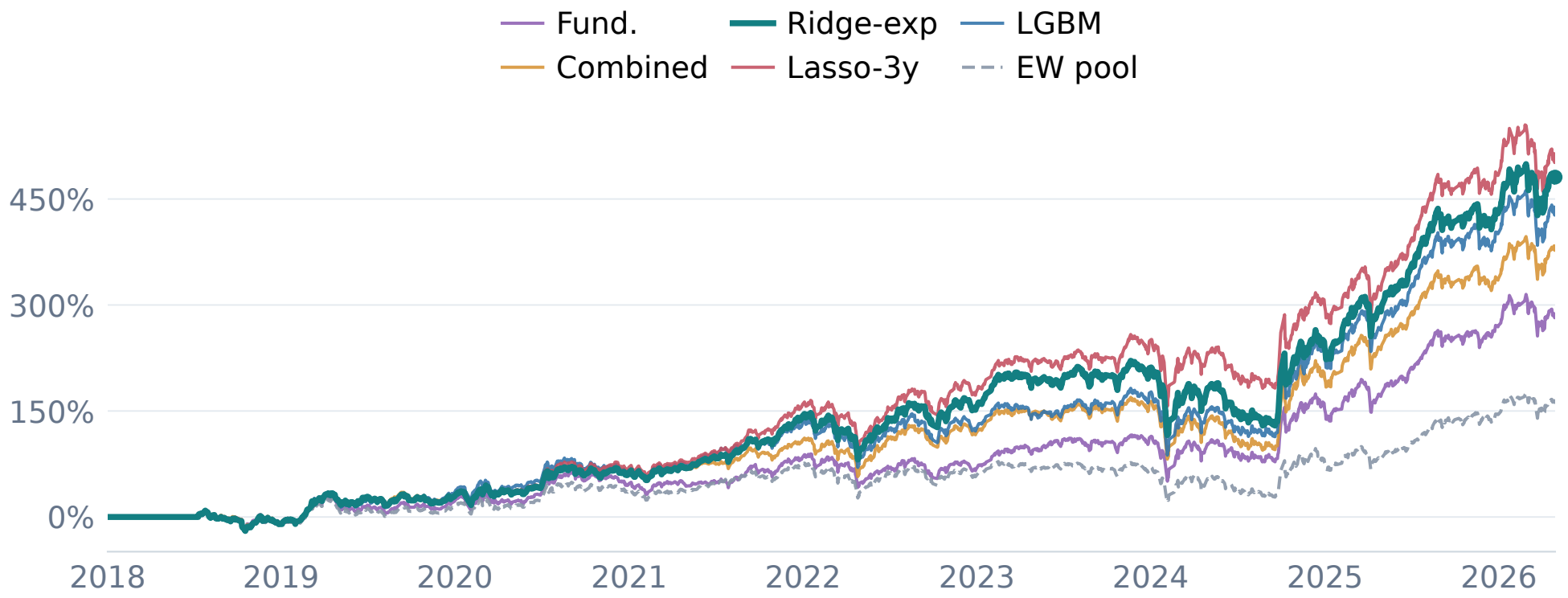
Equal-weight common stock pool
Same cadence and cash constraints
Same restrictions and costs
CSI500 is an auxiliary reference

Commission	Slippage	Sell stamp tax
3 bp per side	5 bp per side	10 → 5 bp on 28 Aug 2023

Twelve sourced company events use explicit claims and settlement dates.

Gross absolute return

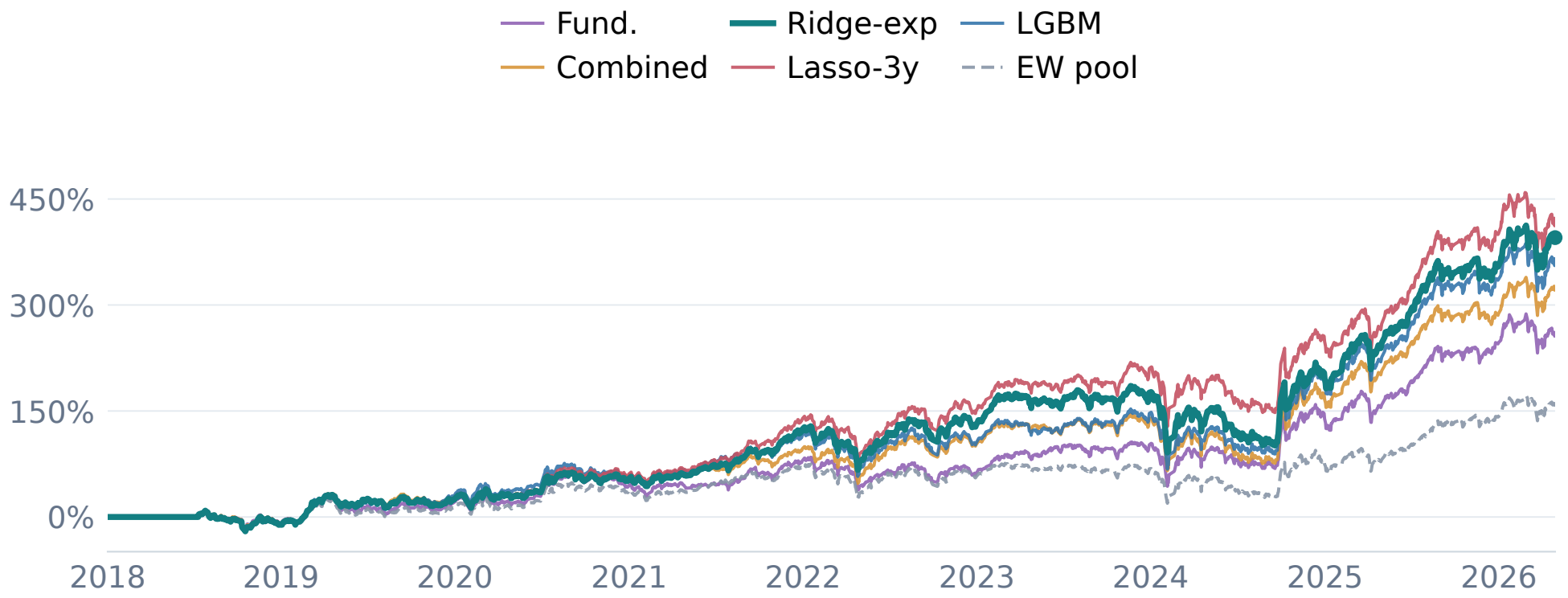
2 Jan 2018 – 29 Apr 2026



Full gross: Lasso-3y 508.00%; Ridge-exp 480.94%.

Net absolute return

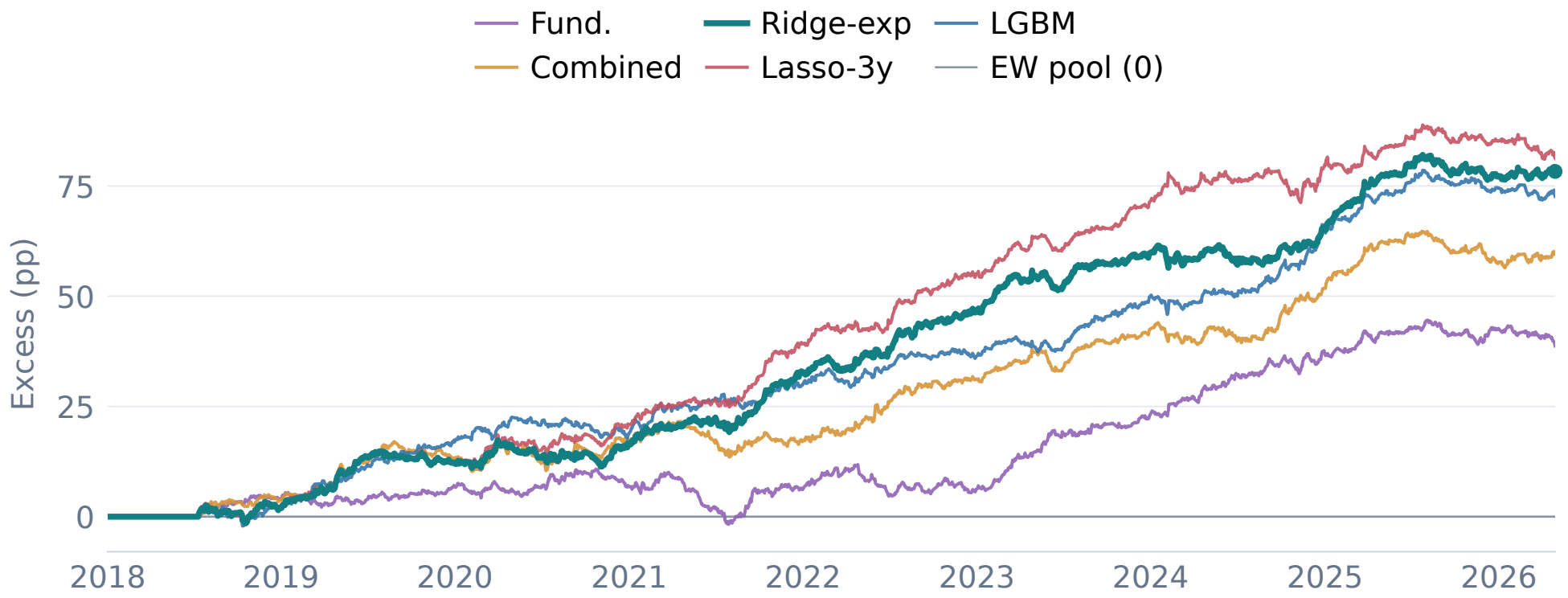
2 Jan 2018 – 29 Apr 2026



Full net: Lasso-3y 417.56%; Ridge-exp 395.35%.

Gross excess return

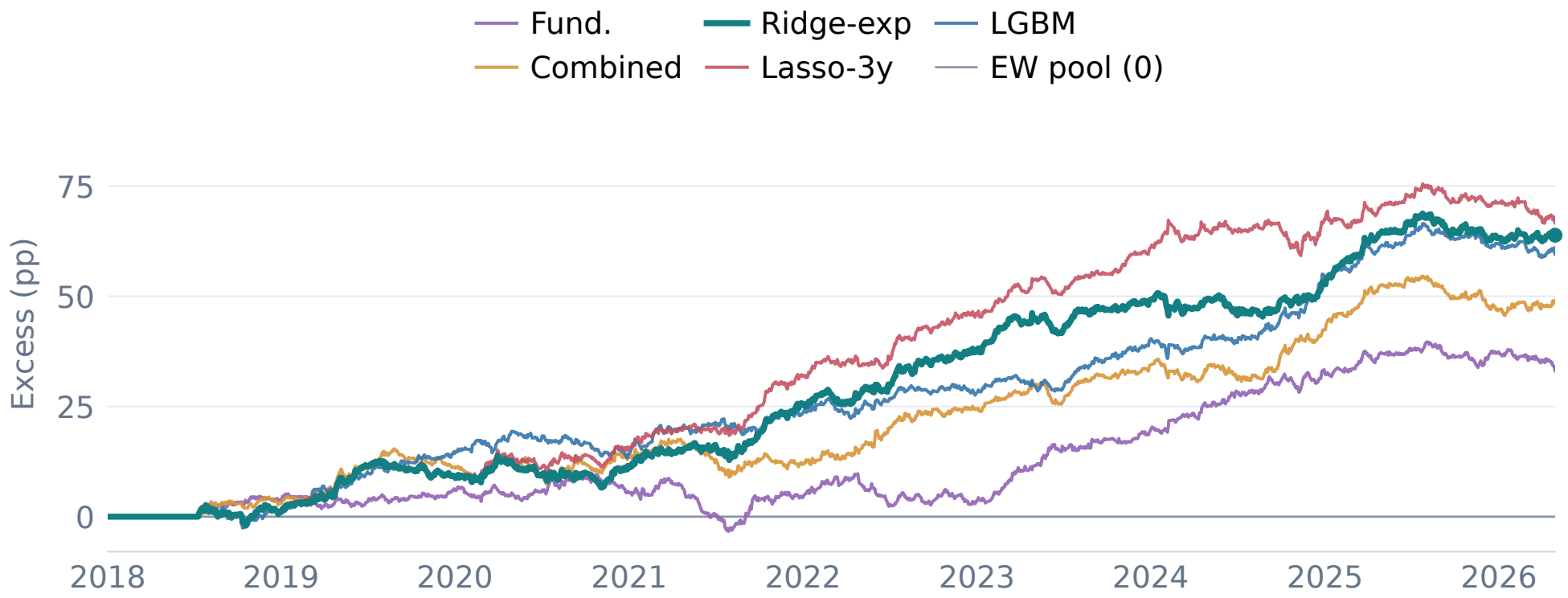
2 Jan 2018 – 29 Apr 2026 · Daily differences versus the gross pool (pp)



The pool's own excess is zero; absolute wealth uses a different calculation.

Net excess return

2 Jan 2018 – 29 Apr 2026 · Daily differences versus the net pool (pp)

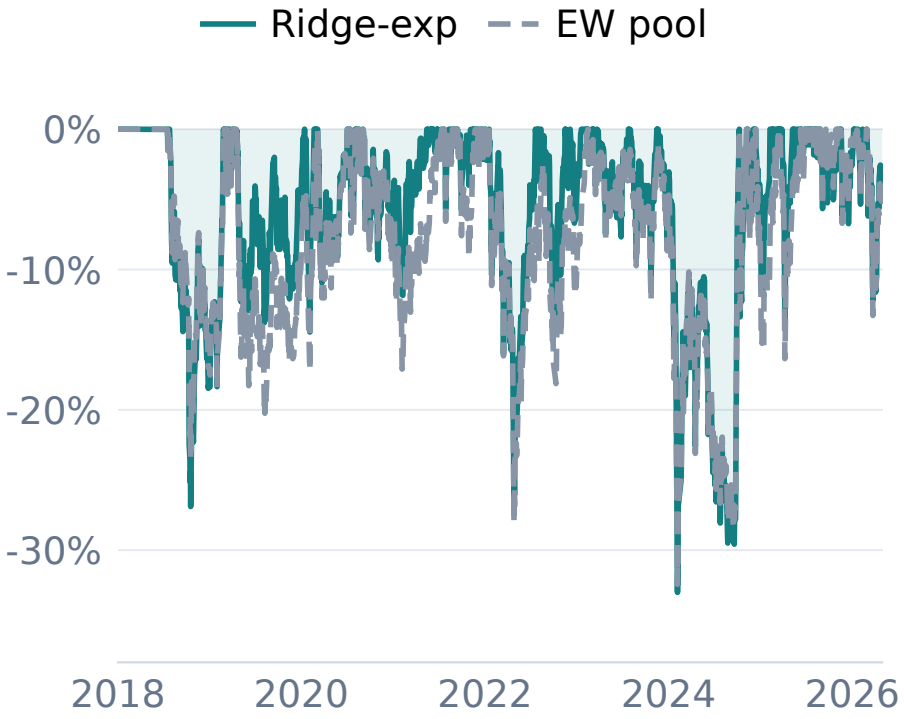


Full net excess: Lasso-3y +66.64 pp; Ridge-exp +63.82 pp.

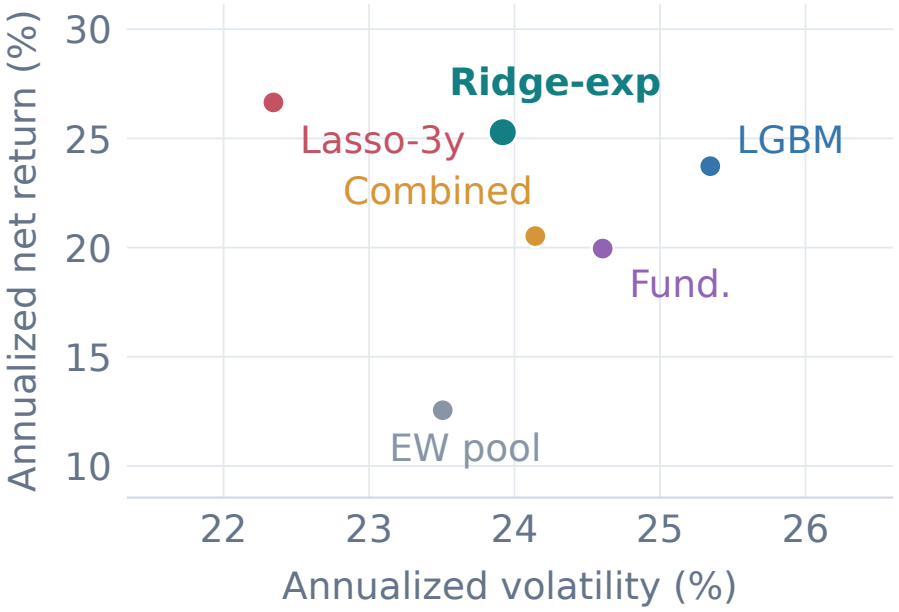
Return comes with drawdown risk

Same published versions · Full-history drawdown; 2021–2025 return / volatility

DRAWDOWN



RETURN / RISK



Risk scatter: Ridge-exp versus Lasso-3y; auxiliary Ridge-3y control is not plotted.

Monthly net excess

Ridge-exp · Sum of daily net differences within each month, in pp

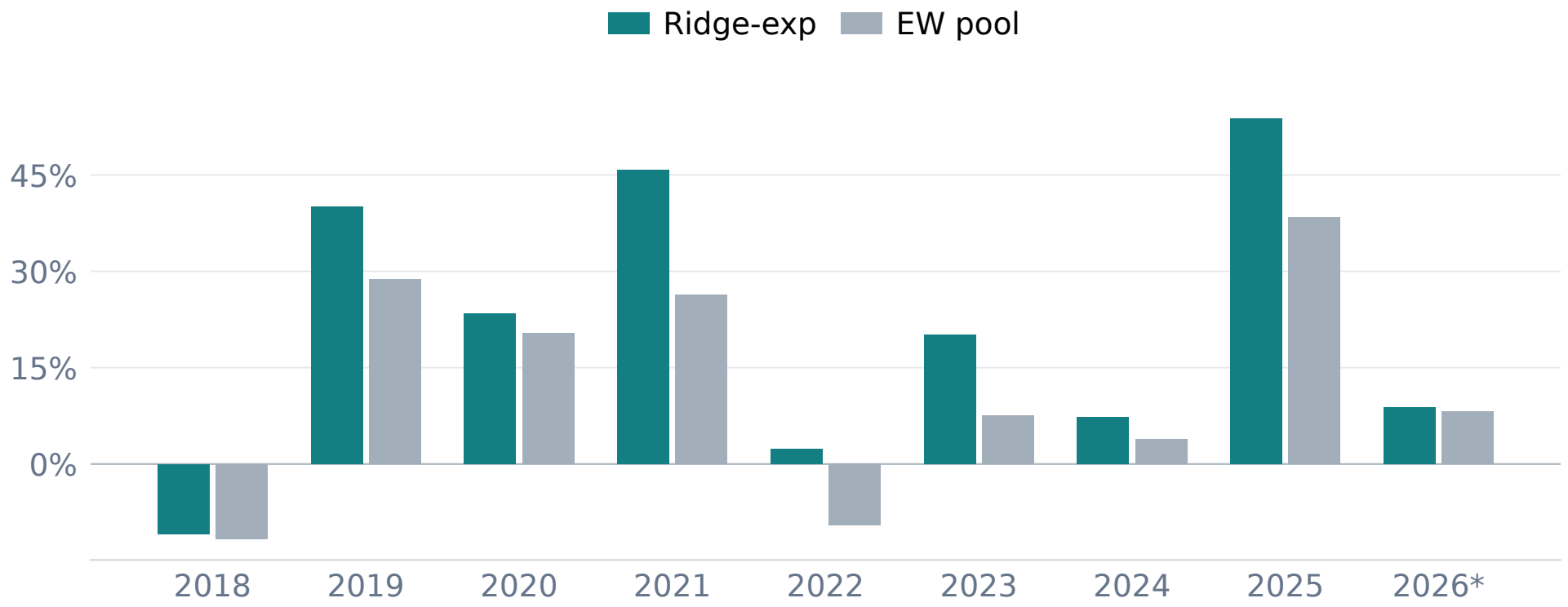
	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
2018	0.0	0.0	0.0	0.0	0.0	0.0	+2.2	-1.2	-0.6	-0.1	+1.9	-1.2
2019	+1.7	+0.7	+1.1	+3.0	+1.4	+2.6	+1.0	-1.2	-0.6	-0.1	-0.5	-1.0
2020	+0.1	+0.1	+4.5	-1.7	-1.6	-1.1	-0.7	+0.8	+0.5	-2.3	+2.5	+1.0
2021	+2.7	+1.3	-1.0	+1.4	-0.3	+0.3	-2.8	+2.9	+3.1	+4.0	0.0	+2.5
2022	+2.1	+0.4	-2.0	+1.9	+1.1	+0.4	+3.8	+2.4	+0.1	+0.6	+0.3	+1.5
2023	+2.1	+3.1	+1.6	+0.6	-1.1	-2.0	+4.7	+0.2	+0.1	+0.2	+1.4	+0.4
2024	+0.5	-1.6	-0.7	+2.3	+0.2	-3.7	+0.7	-0.1	+2.5	+0.2	+0.8	+2.8
2025	+3.7	+2.4	+3.2	+2.5	+0.1	+1.5	+1.5	-0.8	-2.6	+0.1	-0.4	-0.9
2026*	+0.2	-0.3	+0.1	+0.5	—	—	—	—	—	—	—	—



2026 ends on 29 April; later months remain unobserved.

Calendar-year net return

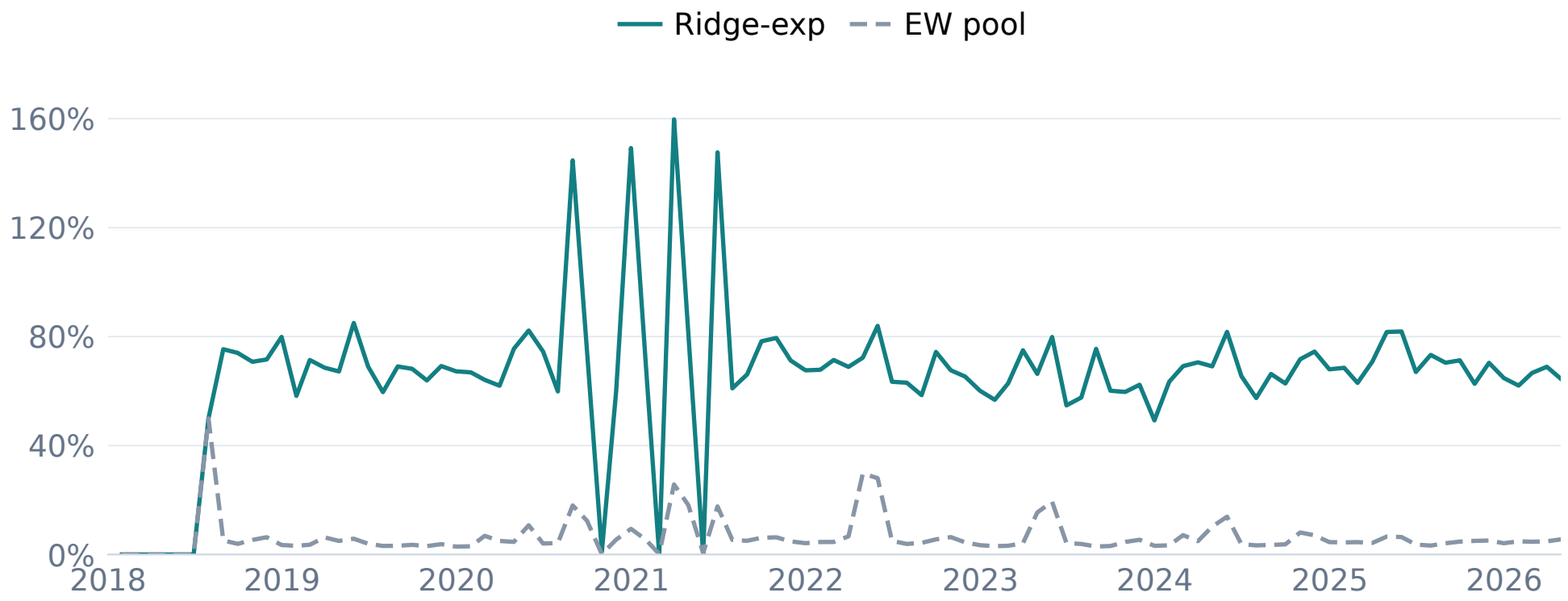
Ridge-exp and the matched pool · Compounded within each calendar year



The 2026 bar covers observations only through 29 Apr 2026.

Monthly portfolio turnover

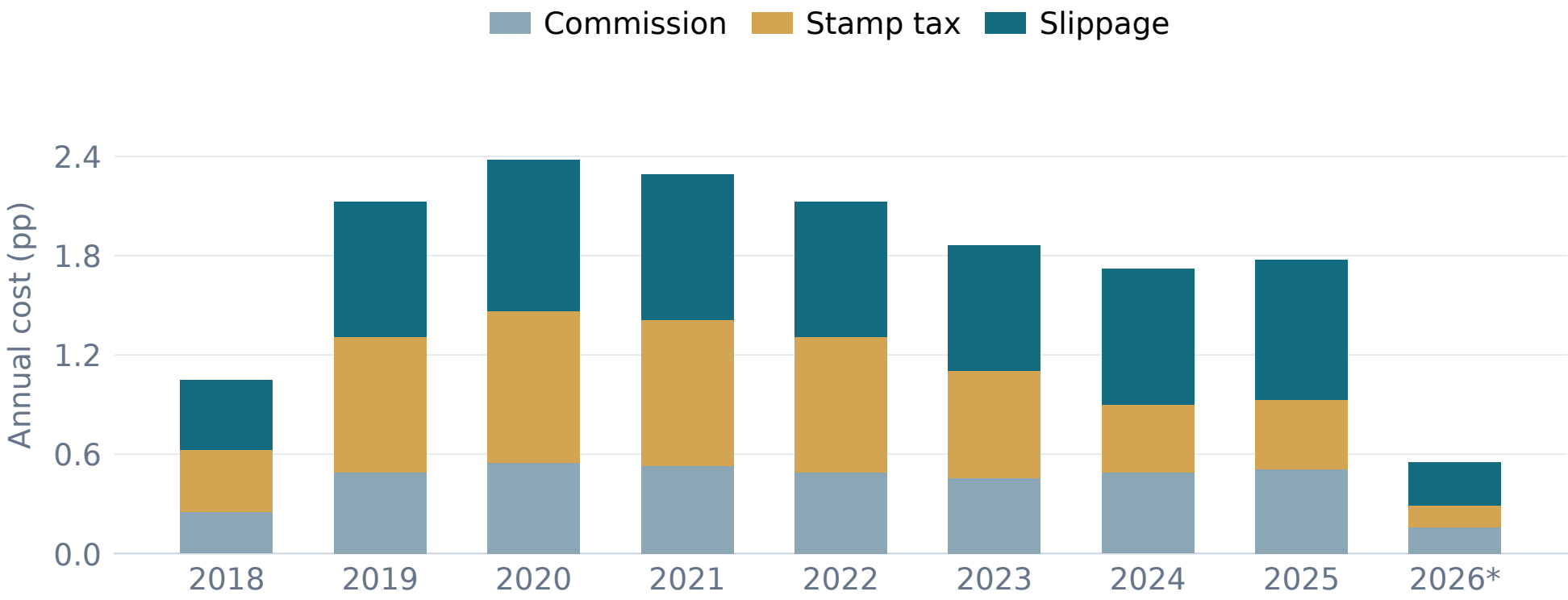
Half of traded notional divided by opening wealth; summed within month



Twenty-session targets retain blocked-order retries and company-event trades.

Trading costs by calendar year

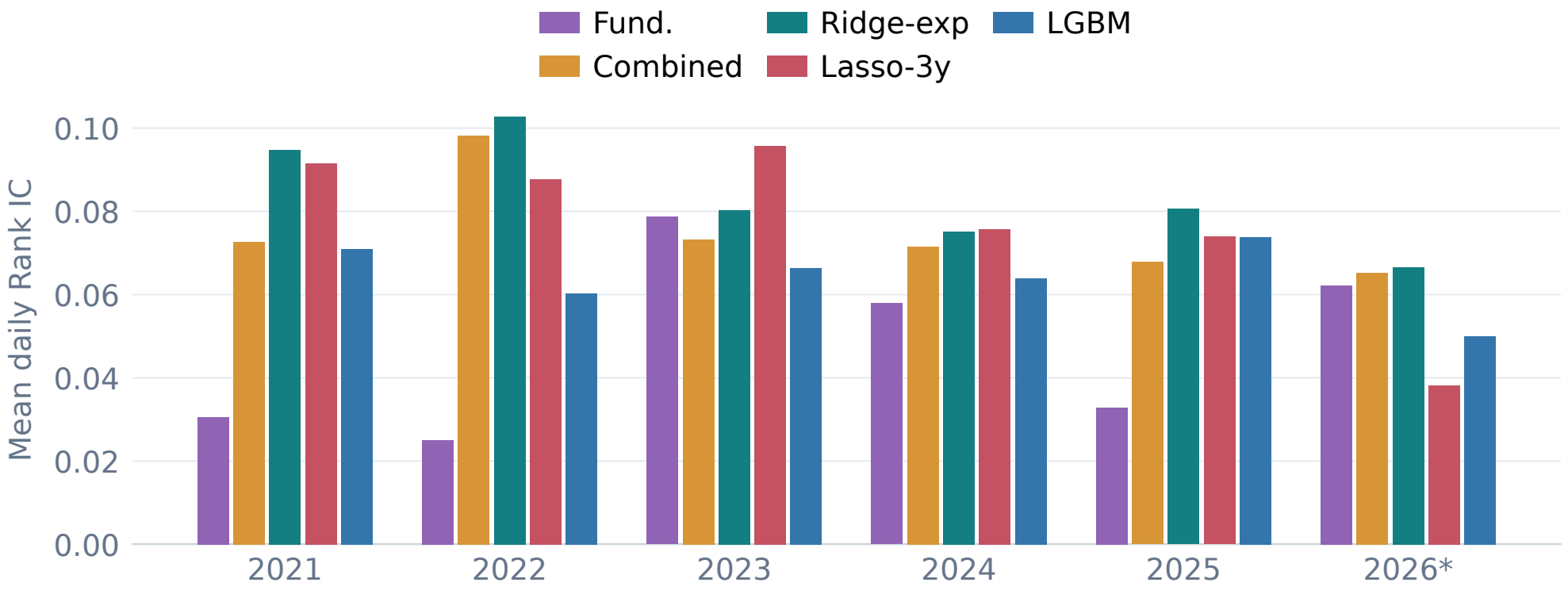
Annual sums of daily cost / opening NAV (pp); 2026 through 29 April



Cost-to-NAV sums and compounded terminal wealth measure different quantities.

Cross-sectional prediction quality

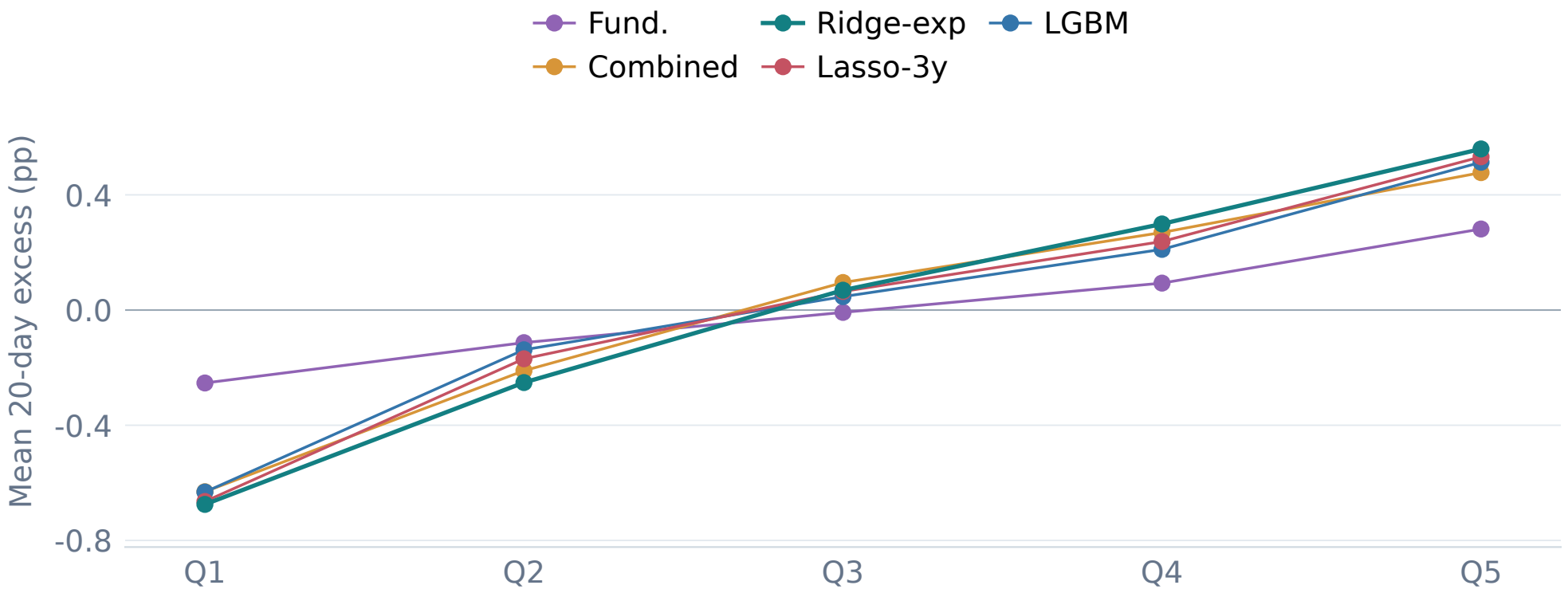
Annual mean Spearman IC · Mature labels; 2026 signals through 30 March



Full-rank prediction quality can coexist with weak top-ranked returns.

Score quintiles: 2021–2025

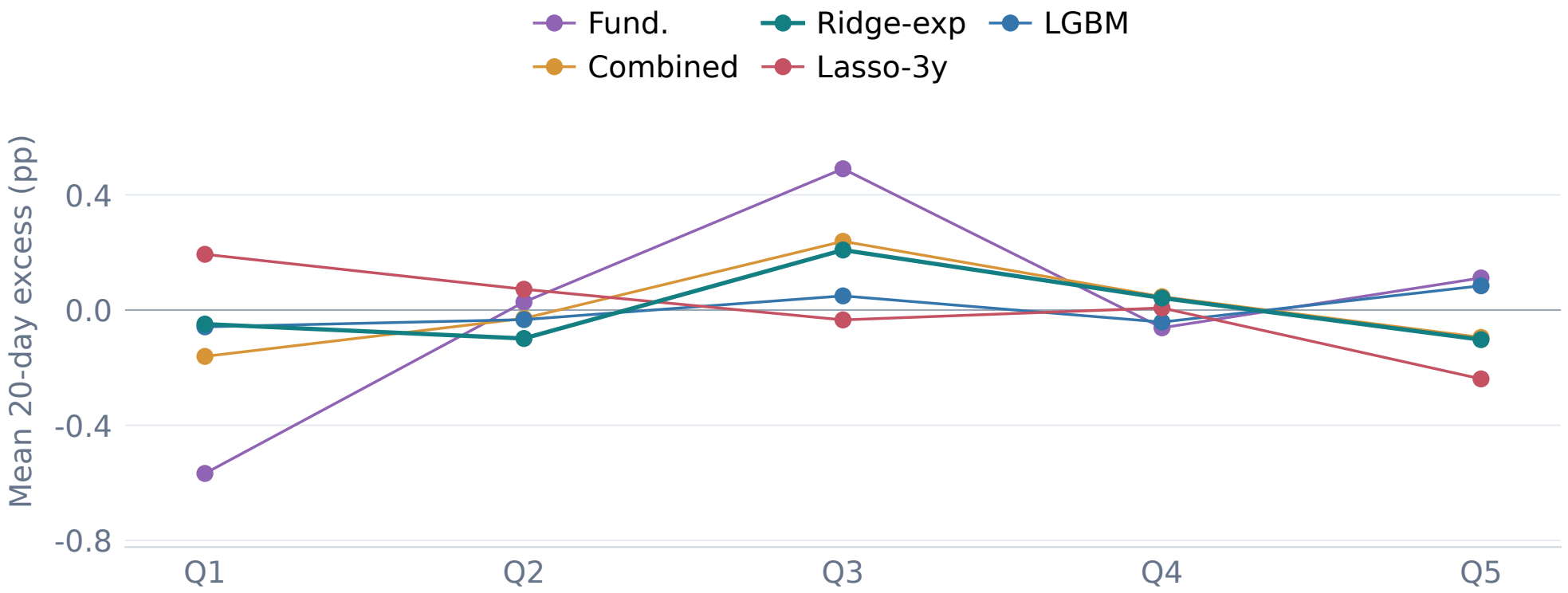
Mean 20-session same-pool excess before costs, in pp; overlapping cohorts



Higher development scores associate with higher cohort excess.

Score quintiles: 2026

55 mature signal dates; last signal 30 Mar 2026; same-pool excess, pp



The latest-year relationship is weak and non-monotone.

Lasso-3y: sparse model, matched experiment

8 inputs · 3-year rolling history · demeaned return target · EMA20

Fit year	Alpha	Nonzero / 8	Validation
2021	3.41e-05	7	2020
2022	0.000199	8	2021
2023	1.47e-09	8	2022
2024	0.000455	4	2023
2025	1.55e-09	8	2024
2026	0.000411	4	2025

Net return: 2021–2025 211.46%; 2026 3.35%.

144 dev tests · Ridge-3y control 223.12% · original Lasso 128.45%.
2026: selected 3.35% vs original Lasso 7.52%; no reselection.

Conclusion

Historical evidence, with clear limits on persistence and implementation

1 Full net: Lasso-3y 417.56%; Ridge-exp 395.35%.

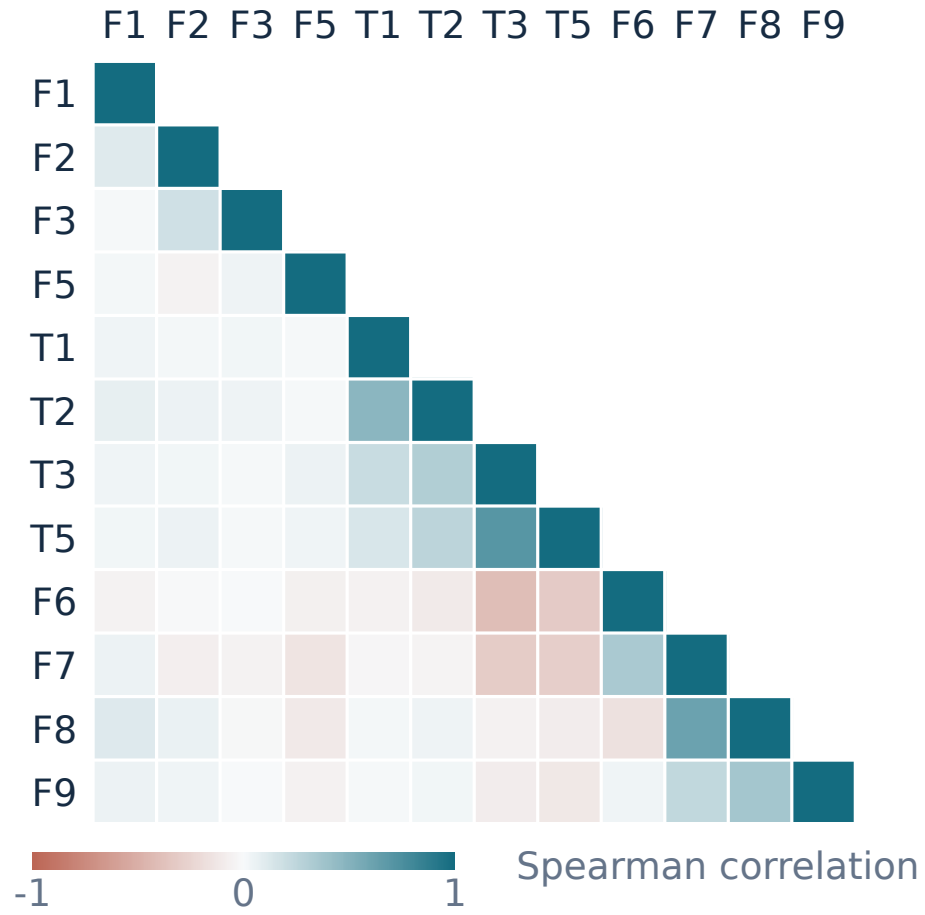
2 Development: Lasso-3y 211.46%, Ridge-exp 195.66%; 2026: 3.35% vs 8.80%.

3 Supplier revisions, event timing and trading assumptions limit the interpretation.

Questions

Appendix A | Factor overlap

Same-date Spearman correlations of the actual annual model inputs



12 used inputs

2021–2025

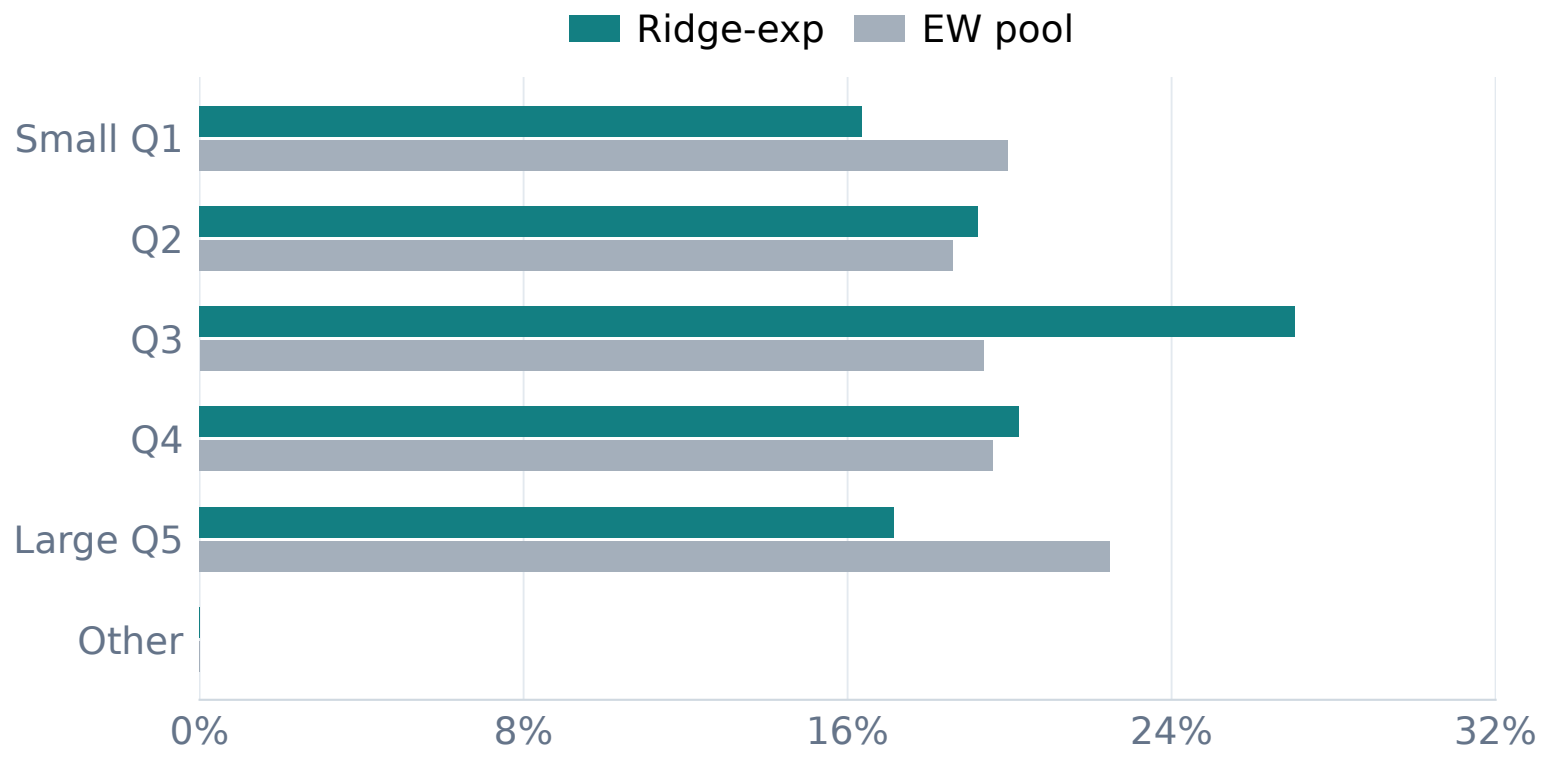
Every fifth session

Historical common pool

Correlation describes signal overlap.

Appendix B | Actual capital weights

29 Apr 2026 · Net-NAV-normalized holdings, cash and claims



Middle-cap Q3: Ridge 27.0%, pool 19.4%; held capital remains style-exposed.

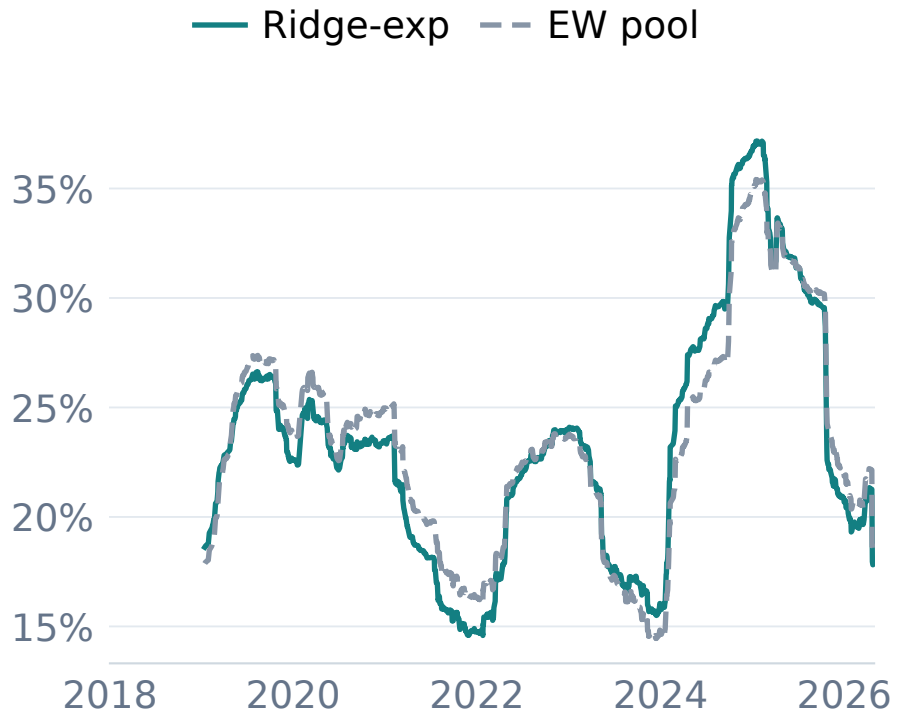
Appendix C | Rolling performance

252-session windows; complete observed history

NET EXCESS (pp)



ANNUALIZED VOLATILITY (%)



Rolling observations overlap; initial incomplete windows stay missing.